Literature-Implied Partial Subcase for arXiv:2004.04805

FA/Banach run fa_banach_001

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Status

This is a `literature_implied_answer (partial subcase)` packet. It does not claim a new solution to the equi-coarse embedding problem in Baudier–Lancien–Motakis–Schlumprecht [1]. It records that a later theorem of Fovelle [2] rules out the stronger equi-Lipschitz embeddings of the Hamming graphs into the particular space $T^*(T^*)$.

Source Problem

Baudier–Lancien–Motakis–Schlumprecht ask whether the Hamming graphs fail to equi-coarsely embed into every reflexive asymptotic-subsequential-c_0 space. They single out the space $T^*(T^*)$:

do the Hamming graphs fail to equi-coarsely embed into $T^*(T^*)$?

They had already proved that, if such embeddings exist, they cannot be of the canonical array/tree forms described earlier in their paper.

Supporting Literature

Fovelle later proves that $T^*(T^*)$ does not have property HC_∞, but does have property $\mathrm{HFC}_{p,d}$ for every $p \in (1,\infty)$. The same paper notes that these finite-p concentration properties prevent equi-Lipschitz embeddings of the Hamming graphs.

Definitions and Identification

Write $H_k^\omega = ([\mathbb{N}]^k, d_H)$ for the k-dimensional Hamming graph. A family $(H_k^\omega)_k$ equi-Lipschitz embeds into a Banach space Y if there are constants $A, B > 0$ and maps $f_k : H_k^\omega \to Y$ such that

$$A\, d_H(\bar{m}, \bar{n}) \leq \|f_k(\bar{m}) - f_k(\bar{n})\| \leq B\, d_H(\bar{m}, \bar{n})$$

for all k and all $\bar{m}, \bar{n} \in [\mathbb{N}]^k$.

For $p \in (1,\infty)$, property $\mathrm{HFC}_{p,d}$ says that there is a constant λ such that, for every k and every Lipschitz map $f : ([\mathbb{N}]^k, d_H) \to Y$, there is an infinite $\mathbb{M} \subseteq \mathbb{N}$ satisfying

$$\|f(\bar{m}) - f(\bar{n})\| \leq \lambda \left(\sum_{j=1}^{k} \alpha_j^p \right)^{1/p}$$

for all $\bar{m}, \bar{n} \in [\mathbb{M}]^k$, where α_j is the directional Lipschitz constant obtained by changing only the jth coordinate.

Idea of the Proof

The source problem is about coarse embeddings, whose compression may grow sublinearly. Fovelle's finite-p concentration theorem is strong enough to rule out the linear-scale case. Indeed, an equi-Lipschitz embedding would force disjoint k-sets to be separated by order k, while $\mathrm{HFC}_{p,d}$ forces a large sub-Hamming graph to have diameter at most order $k^{1/p}$. Since $p > 1$, these estimates are incompatible for large k.

Theorem 1. *The Hamming graphs $(H_k^\omega)_{k\geq 1}$ do not equi-Lipschitz embed into $T^*(T^*)$.*

Proof. Choose any $p \in (1, \infty)$. By Fovelle's corollary [2], $T^*(T^*)$ has property $\mathrm{HFC}_{p,d}$; let λ be a constant witnessing this property.

Assume, for contradiction, that there are constants $A, B > 0$ and maps $f_k : H_k^\omega \to T^*(T^*)$ such that

$$A\, d_H(\bar{m}, \bar{n}) \leq \|f_k(\bar{m}) - f_k(\bar{n})\| \leq B\, d_H(\bar{m}, \bar{n})$$

for every k and all $\bar{m}, \bar{n} \in [\mathbb{N}]^k$.

For each k, every directional Lipschitz constant α_j of f_k is at most B, because changing one coordinate changes the Hamming distance by one. Applying $\mathrm{HFC}_{p,d}$ to f_k, we obtain an infinite $\mathbb{M} \subseteq \mathbb{N}$ such that

$$\|f_k(\bar{m}) - f_k(\bar{n})\| \leq \lambda \left(\sum_{j=1}^{k} B^p \right)^{1/p} = \lambda B k^{1/p}$$

for all $\bar{m}, \bar{n} \in [\mathbb{M}]^k$.

Since $\mathbb{M}$ is infinite, choose disjoint $\bar{m}, \bar{n} \in [\mathbb{M}]^k$. Then $d_H(\bar{m}, \bar{n}) = k$, so the lower Lipschitz estimate gives

$$Ak \leq \|f_k(\bar{m}) - f_k(\bar{n})\| \leq \lambda B k^{1/p}.$$

This is impossible for all sufficiently large k, because $p > 1$. Hence no such equi-Lipschitz family exists. □

Verification and Limitations

The packet uses no computational verifier. The proof is a direct consequence of Fovelle's concentration property and the definitions of equi-Lipschitz embedding and Hamming distance.

The limitation is essential: this proves only the failure of equi-Lipschitz embeddings into $T^*(T^*)$. It does not rule out equi-coarse embeddings with sublinear compression, which is the question asked in [1].

Novelty and Search Bounds

This is not an originality claim. Before packaging, the run's cheap indexes were searched for `2004.04805`, Hamming graphs, Johnson graphs, $T^*(T^*)$, asymptotic-subsequential-c_0, and HFC keywords. No prior packet or attempt for this exact source problem or subcase was found. Local corpus and web searches for the Hamming/Johnson and $T^*(T^*)$ phrases found the source paper and later concentration-property papers, with arXiv:2106.04297 providing the supporting theorem used here.

Human Review Recommendation

Review as a provenance/status packet. The main points to check are that Fovelle's $\mathrm{HFC}_{p,d}$ convention uses the same Hamming graph metric as the source paper and that the finite-p concentration argument above correctly rules out equi-Lipschitz embeddings. Do not count this as a full answer to the source paper's equi-coarse problem.

References

[1] F. Baudier, G. Lancien, P. Motakis, and Th. Schlumprecht, *The geometry of Hamming-type metrics and their embeddings into Banach spaces*, arXiv:2004.04805.

[2] A. Fovelle, *Hamming graphs and concentration properties in non-quasi-reflexive Banach spaces*, arXiv:2106.04297.

5. Final remarks and open problems

Although we do not know whether or not the Hamming graphs equi-coarsely embed into $T^*(T^*)$ we now understand that if such embeddings were to exist they would not be of any of the canonical types that we have described in Proposition 4.6 and Remark 4.7.

Problem 5.1. *Is it true that the Hamming graphs do not equi-coarsely embed into any reflexive asymptotic-subsequential-c_0 space? In particular, is it true that the Hamming graphs do not equi-coarsely embed into $T^*(T^*)$?*

The class of asymptotic-subsequential-c_0 spaces is a new one. This is not surprising as even proving that $T^*(T^*)$ has this property is non-trivial and the motivation for defining this property presented itself only now. A more general theorem can be shown, albeit with a more technical proof.

Theorem 5.2. *The T^*-sum of any sequence of C-asymptotic-c_0 spaces for a uniform constant C is asymptotic-subsequential-c_0.*

Such examples contain many asymptotic-c_0 subspaces.

Problem 5.3. *Let X be an infinite dimensional asymptotic-subsequential-c_0 space. Does X contain an infinite dimensional asymptotic-c_0 subspace?*

Next we describe a particular Banach space and some of its properties which are interesting regarding the study of certain asymptotic properties under a metrical scope. This example is based on the original idea of Szlenk in [Szl68]. It is also related to [OSch02, Example 4.2]. For $1 < p < \infty$ and $1 \le q \le \infty$ we can construct a reflexive Banach space $X_p^{q,\omega}$ with the following property: all asymptotic models generated by normalized weakly null arrays in $X_p^{q,\omega}$ are isometrically equivalent to the unit vector basis of ℓ_p, yet ℓ_q^k is (isometrically) in the k-th asymptotic structure of $X_p^{q,\omega}$ for every $k \in \mathbb{N}$. Therefore a statement which is analogous to Theorem 3.5 for ℓ_p, $1 < p < \infty$, cannot be true.

The construction of the space $X_p^{q,\omega}$ that we are about to describe is based on the idea of Szlenk from [Szl68], and is somewhat similar to [OSch02, Example 4.2]. Fix $1 < p < \infty$ and $1 \le q \le \infty$ and define by induction a sequence of spaces $(X_p^{q,k})_k$ as follows. Set $X_p^{q,0} = \mathbb{R}$ and then set $X_p^{q,k} = \mathbb{R} \oplus_q \ell_p(X_p^{q,k-1})$. Finally, define $X_p^{q,\omega} = (\oplus_{k=0}^\infty X_p^{q,k})_p$. Each space $X_p^{q,k}$ is reflexive and so is $X_p^{q,\omega}$. The fact that all asymptotic models generated by normalized weakly null arrays in $X_p^{q,\omega}$ are isometrically equivalent to the unit vector basis of ℓ_p can be proved as follows. Use Proposition 2.5 to show by induction that for all $k \in \mathbb{N}$ all the asymptotic models generated by normalized weakly null arrays in $X_p^{q,k}$ are isometrically equivalent to the ℓ_p-unit vector basis, and use Proposition 2.5 one more time to obtain the same conclusion for $X_p^{q,\omega}$. We now turn to the statement about the asymptotic structure of $X_p^{q,\omega}$.

Proposition 5.4. *Let $p \in (1,\infty)$ and $q \in [1,\infty]$. For every $k \in \mathbb{N} \cup \{0\}$ the space $X_p^{q,k}$ contains a normalized weakly null tree $(x_{\bar{m}} : \bar{m} \in [\mathbb{N}]^{\le k})$, all branches of which are isometrically equivalent to the unit vector basis of ℓ_q^k.*

Proof. For $k = 0$ pick a norm-one vector $x_\emptyset$ in $X_p^{q,0} = \mathbb{R}$. Let now $X_p^{q,k} = \mathbb{R} \oplus_q \ell_p(X_p^{q,k-1})$ and let for each $i \in \mathbb{N}$ $(x_{\bar{m}}^{(i)} : \bar{m} \in [\mathbb{N}]^{\le k-1})$ be a normalized weakly null tree in the i'th copy of $X_p^{q,k-1}$ all branches of which are isometrically equivalent to the unit vector basis of ℓ_q^{k-1}. Take $x_\emptyset$ to be a norm-one vector in $X_p^{q,k}$ that resides in $\mathbb{R}$ (the left part of the sum $X_p^{q,k} = \mathbb{R} \oplus_q \ell_p(X_p^{q,k-1})$) and for $1 \le n \le k$ and $\bar{m} = \{m_1, \ldots, m_n\}$ define $x_{\bar{m}} = x^{m_1}_{\{m_2-m_1,\ldots,m_n-m_1\}}$ (in particular, for $\bar{m} = \{m\}$, $x_{\bar{m}} = x^m_\emptyset$). $\square$

Figure 1: Source evidence from arXiv:2004.04805, page 26: final remarks and Problem 5.1 asking about equi-coarse embeddings of Hamming graphs into reflexive asymptotic-subsequential-c_0 spaces and into $T^*(T^*)$.

for any $2k \leqslant j_1 < \cdots < j_{2k}$ and any $a_1, \ldots, a_{2k} \in [-1, 1]$, we have

$$\left| \left\| \sum_{i=1}^{2k} a_i x_{j_i}^{(i)} \right\| - \left\| \sum_{i=1}^{2k} a_i e_i \right\| \right| < \delta. \tag{1}$$

Define now $f(\overline{m}) = \frac{1}{2} \sum_{i=1}^{k} x_{m_i}^{(i)}$ for $\overline{m} = (m_1, \ldots, m_k) \in [\mathbb{N}]^k$. Note that f is 1-Lipschitz for the metric $d_{\mathbb{H}}$.
Let $\mathbb{M} \in [\mathbb{N}]^\omega$, $\overline{n}, \overline{m} \in [\mathbb{M}]^k$ such that $((n_1, \ldots, n_k), (m_1, \ldots, m_k)) \in I_k(\mathbb{M})$ and $m_1, n_1 > 2k$. Using the Hahn-Banach Theorem, we can find $x^* \in S_{X^*}$ such that :

$$x^* \left(\sum_{i=1}^{k} x_{m_i}^{(i)} - \sum_{i=1}^{k} x_{n_i}^{(i)} \right) = \left\| \sum_{i=1}^{k} x_{m_i}^{(i)} - \sum_{i=1}^{k} x_{n_i}^{(i)} \right\|.$$

Using equation (1), we deduce that :

$$\| f(m_1, \ldots, m_k) - f(n_1, \ldots, n_k) \| \geqslant \frac{1}{2} \lambda_{2k} - \frac{\delta}{2} = \frac{\lambda_{2k}}{4} > \lambda.$$

This contradicts the assumption on X. The result follows. □

Theorem 1.1 and Corollary 3.4 provided us with examples of non-quasi-reflexive Banach spaces having property HC_p, for $p \in (1, \infty)$. In order to obtain a similar result with $p = \infty$, it seems natural to consider a T^*-sum of spaces with λ-HFC_∞, $\lambda > 0$. However, as a direct consequence of Theorem 4.5, we get the following corollary.

Corollary 4.6. *The space $T^*(T^*)$, where T^* is the original Banach space constructed by Tsirelson in* [37], *does not have property HC_∞ although it has property $HFC_{p,d}$ for every $p \in (1, \infty)$.*

Before proving this corollary, let us recall some properties of the space T^*. First, it is reflexive, so we will denote its dual by T. This space T^* has a normalized, shrinking, 1-unconditional basis $(e_n)_{n=1}^\infty$. Let us denote by $(e_n^*)_{n \in \mathbb{N}} \subset T$ the coordinate functionals of $(e_n)_{n \in \mathbb{N}}$. For an element $x = \sum_{n=1}^\infty e_n^*(x) e_n \in X$, let us denote by $\mathrm{Supp}(x)$ the support of x, *i.e*, the subset of integers n such that $e_n^*(x) \neq 0$. The space T satisfies the following (see [37] and [14]): for every $(x_i)_{i=1}^n \subset T$ with $(\mathrm{Supp}(x_i))_{i=1}^n$ increasing, $\|x_i\| = 1$, and $\mathrm{Supp}(x_1) \subset [k+1, \infty)$, we have

$$\forall (a_i)_{i=1}^n \subset \mathbb{R}, \left\| \sum_{i=1}^n a_i x_i \right\| \geqslant \frac{1}{2} \sum_{i=1}^n |a_i|.$$

Proof of Corollary 4.6. The lack of property HC_∞ is a direct consequence of Lemma 2.7 of [4] (asserting that $T^*(T^*)$ is not asymptotic-c_0) and Theorem 4.5 above. The fact that $T^*(T^*)$ has property $\mathrm{HFC}_{p,d}$, for every $p \in (1, \infty)$, can be deduced from Theorem 5.9 of [11], applied with $p = 1$, $X_n = T^*$ for all $n \in \mathbb{N}$ and $\mathfrak{E} = T^*$. Indeed, if we let

$$\mathrm{p}(X) = \inf \left\{ q \geqslant 1;\ X \text{ is } p\text{-AUSable}, \frac{1}{p} + \frac{1}{q} = 1 \right\}$$

for a given Banach space X, Theorem 5.9 [11] asserts that $\mathrm{p}(T^*(T^*)) = 1$. As $T^*(T^*)$ is reflexive, the result follows from Theorem 6.1 [22] of Kalton and Randrianarivony (see the remark after Definition 2.5). □

As property HC_∞ prevents the equi-coarse embeddability of the Hamming graphs, we can therefore ask the following :

Figure 2: Supporting evidence from arXiv:2106.04297, page 17: the corollary stating that $T^*(T^*)$ has property $\mathrm{HFC}_{p,d}$ for every $p \in (1, \infty)$.